```
-- Q1 Display the names, email addresses, and credit limits of all customers.
select
    customername,
    customeremail,
    customercreditlimit
from customer;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customername	customeremail	customercreditlimit
Flor Stone	flor.stone@raytheon.com	5000
Lavera Emerson	lavera.emerson@plainsallamerican.com	5000
Fern Head	fern.head@usfoods.com	1200
Shyla Ortiz	shyla.ortiz@abbvie.com	2400
Jeni Levy	jeni.levy@centene.com	1200
Matthias Hannah	matthias.hannah@chs.net	500
Matthias Cruise	matthias.cruise@alcoa.com	500
Meenakshi Mason	meenakshi.mason@internationalpaper.com	3600
Christian Cage	christian.cage@emerson.com	2400
Charlie Sutherland	charlie.sutherland@up.com	3600
Charlie Pacino	charlie.pacino@amgen.com	500

```
-- Q2 Find all customers whose credit limit is greater than 3000.
select *
from customer
where customercreditlimit > 3000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CustomerID	CustomerName	CustomerEmail	CustomerPhone	CustomerAddress	CustomerCreditLimit
2301	Flor Stone	flor.stone@raytheon.com	13171234104	2904 S Salina St	5000
2302	Lavera Emerson	lavera.emerson@plainsallamerican.com	13171234111	5344 Haverford Ave, Philadelphia	5000
2308	Meenakshi Mason	meenakshi.mason@internationalpaper.com	15652788999	1831 No Wong, Peking	3600
2310	Charlie Sutherland	charlie.sutherland@up.com	17887873099	Piazza Cacchiatore 23, San Giminiano	3600
2313	Daniel Costner	daniel.costner@staples.com	18779099109	1597 Legend St, Mysore, Kar	3700
2314	Dianne Derek	dianne.derek@danaher.com	18717872090	1606 Sangam Blvd, New Delhi,	5000
2329	Willie Barrera	willie.barrera@starbucks.com	18990924161	Po Box 2152, Buffalo, NY	5000
2331	Marlene Odom	marlene.odom@cummins.com	12650900181	1617 Crackers St, Bangalore - India, Kar	5000
2335	Cora Calhoun	cora.calhoun@kimberly-clark	16782091010	1656 Veterans Rd, Chennai, Tam	3700
2340	Amira Macdonald	amira.macdonald@fluor.com	17890988761	1668 Chong Tao, Beijing,	5000
2347	Verena Hopper	verena.hopper@generalmills.com	6710191725	612 Jefferson Ave, Scranton, PA	3600

```
-- Q3 Display the details of employees whose job title is 'Programmer' and who were hired after January 2016.
select *
from employee
where employeejobtitle ='programmer' and employeehiredate > '2016-01-31';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EmployeeID	EmployeeName	EmployeeEmail	EmployeePhone	employeehiredate	EmployeeJobTitle	WarehouseID
6323	Nathan Cox	nathan.cox@example.com	5904234568	2016-05-21	Programmer	115
6324	Bobby Torres	bobby.torres@example.com	5904235567	2016-02-07	Programmer	116
6325	Charles Ward	charles.ward@example.com	5904234560	2016-02-05	Programmer	117
6326	Gabriel Howard	gabriel.howard@example.com	59042345692	2016-06-25	Programmer	118
6432	Beak Kimberly	BeakKimberly@gmail.com	17925003303	2016-07-21	Programmer	224
6643	Rachael Maggie	RachaelMaggie@gmail.com	11594484231	2016-10-02	Programmer	435

```sql
-- Q4 Display all products with their product name, category and list price, sorted by list price from highest to lowest.
select
    productname,
    categoryname,
    productlistprice
from product
order by productlistprice desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

productname	categoryname	productlistprice
G.Skill RipjawsX (F3-1600C9Q-32GXM) DDR3 32...	RAM	78180
Corsair Vengeance LPX (CMK16GX4M2B3000C1...	RAM	14999
G.Skill TridentX (F3-2400C10D-8GTX) DDR3 8GB	RAM	14540
Intel SSDPECME040T401	Storage	7123.66
Western Digital WD2003FZEX	Storage	7123.66
Samsung MZ-V6P512BW	Storage	7123.66
Hitachi HUS724030ALE641	Storage	7123.66
Samsung MZ-V6P2T0BW	Storage	7123.66
Seagate ST10000DM0004	Storage	7123.66
ADATA ASU800SS-512GT-C	Storage	7123.66
Samsung MZ-V6E1T0	Storage	7123.66

Result Grid | Form Editor | Field Types

```sql
-- Q5 Find the 5 most expensive products based on their list price.
select *
from product
order by productlistprice desc
limit 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

ProductID	ProductName	CategoryName	ProductDescription	ProductStandardCost	ProductListPric
P307	G.Skill RipjawsX (F3-1600C9Q-32GXM) DDR3 32...	RAM	Series:RipjawX, Type:DDR3, Capacity:32GB, Fr...	75410	78180
P299	Corsair Vengeance LPX (CMK16GX4M2B3000C1...	RAM	Series:N/A, Type:DDR3, Capacity:16GB, Frequ...	13250	14999
P306	G.Skill TridentX (F3-2400C10D-8GTX) DDR3 8GB	RAM	Series:TridentX, Type:DDR3, Capacity:8GB, Fre...	12578	14540
P253	Western Digital WD2003FZEX	Storage	Series:BLACK SERIES,Type:7200RPM,Capacity:...	7123.66	7123.66
P341	Intel SSDPECME040T401	Storage	Series:DC P3608,Type:SSD,Capacity:4TB,Cach...	7123.66	7123.66

```sql
-- Q6 Display all unique product categories available in the product table.
select
    distinct categoryname
from product;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

categoryname
CPU
Video Card
Storage
Mother Board
RAM

```sql
-- Q7 Find all order details where the order item quantity is greater than 100 and sort them by quantity in descending order.
select *
from orderdetails
where orderitemquantity > 100
order by orderitemquantity desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

OrderDetailsID	ProductID	OrderItemQuantity	PerUnitPrice	OrderStatus	OrderID
399	P399	157	821.99	Shipped	7670
60	P060	150	1314.99	Shipped	7331
316	P316	150	849.99	Pending	7587
17	P017	149	339.99	Shipped	7288
306	P306	149	1199.99	Pending	7577
307	P307	149	2064.99	Canceled	7578
6	P006	148	549.59	Shipped	7277
23	P023	148	47.88	Shipped	7294
263	P263	148	829.89	Shipped	7534
101	P101	147	2200	Canceled	7372
342	P342	147	573.99	Shipped	7613

```sql
-- Q8 Find the 10 products with the highest profit and display their ProductID, ProductName, ProductListPrice, and Profit.
select
    productid,
    productname,
    productlistprice,
    profit
from product
order by profit desc
limit 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

productid	productname	productlistprice	profit
P307	G.Skill RipjawsX (F3-1600C9Q-32GXM) DDR3 32...	78180	2770
P320	A-DATA (AX4U300038G16-SRS) 8GB DDR4	6990	1990
P306	G.Skill TridentX (F3-2400C10D-8GTX) DDR3 8GB	14540	1962
P299	Corsair Vengeance LPX (CMK16GX4M2B3000C1...	14999	1749
P071	PNY VCQP6000-PB	5499.99	1441
P073	PNY VCQM6000-PB	3254.99	749.95
P003	Intel Xeon E5-2698 V3 (OEM/Tray)	2660.72	625.54
P078	AMD 100-505989	2699.99	571.32
P382	AMD 100-505989	2699.99	571.32
P383	PNY VCQK6000-PB	2290.79	550.48

```sql
-- Q9 Display each order along with the customer's name and the order date.
-- Display:
-- OrderID
-- OrderDate
-- CustomerID
-- CustomerName
select
    o.orderid,
    o.orderdate,
    c.customerid,
    c.customername
from customer as c
join orders as o
on c.customerid = o.customerid;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

orderid	orderdate	customerid	customername
7272	2016-11-17	2301	Flor Stone
7273	2017-02-20	2302	Lavera Emerson
7274	2017-01-03	2303	Fern Head
7275	2017-10-15	2304	Shyla Ortiz
7276	2017-04-09	2305	Jeni Levy
7277	2017-08-15	2306	Matthias Hannah
7278	2016-12-01	2307	Matthias Cruise
7279	2015-04-09	2308	Meenakshi Mason
7280	2017-05-11	2309	Christian Cage

```sql
-- Q10 Find the number of orders placed by each customer.
-- Display:
-- CustomerID
-- CustomerName
-- Total Orders
select
    c.customerid,
    c.customername,
    count(o.orderid) as total_order
from customer as c
join orders as o
on c.customerid = o.customerid
group by c.customerid, c.customername;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customerid	customername	total_order
2301	Flor Stone	1
2302	Lavera Emerson	1
2303	Fern Head	1
2304	Shyla Ortiz	1
2305	Jeni Levy	1
2306	Matthias Hannah	1
2307	Matthias Cruise	1
2308	Meenakshi Mason	1
2309	Christian Cage	1
2310	Charlie Sutherland	1

```sql
-- Q11 Display every customer along with the number of orders they have placed, including customers who have not placed any orders.
select
    c.customername,
    count(o.orderid) as num_orders
from customer as c
left join orders as o
on c.customerid = o.customerid
group by c.customerid, c.customername;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customername	num_orders
Flor Stone	1
Lavera Emerson	1
Fern Head	1
Shyla Ortiz	1
Jeni Levy	1
Matthias Hannah	1
Matthias Cruise	1

```sql
-- Q12 Find the total sales generated by each product.
-- Display:
-- ProductID
-- ProductName
-- Total Sales
select
    p.productid,
    p.productname,
    sum(o.orderitemquantity * o.perunitprice) as total_sales
from product as p
join orderdetails as o
on p.productid = o.productid
group by p.productid, p.productname;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

productid	productname	total_sales
P001	Intel Xeon E5-2699 V3 (OEM/Tray)	62038.68
P002	Intel Xeon E5-2697 V3	64478.76
P003	Intel Xeon E5-2698 V3 (OEM/Tray)	73668.08
P004	Intel Xeon E5-2697 V4	108798.72
P005	Intel Xeon E5-2685 V3 (OEM/Tray)	11658.939999999999
P006	Intel Xeon E5-2695 V3 (OEM/Tray)	81339.32
P007	Intel Xeon E5-2697 V2	83818.73
P008	Intel Xeon E5-2695 V4	7588.62

```sql
-- Q13 Find the total quantity of products sold in each product category.
-- Display:
-- CategoryName
-- Total Quantity Sold
select
    p.categoryname,
    count(o.orderitemquantity) as total_quantity
from product as p
join orderdetails as o
on p.productid = o.productid
group by p.categoryname;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

categoryname	total_quantity
CPU	90
Video Card	70
Storage	127
Mother Board	79
RAM	34

```sql
-- Q14 Find the product categories whose total sales exceed 50,000.
-- Display:
-- CategoryName
-- Total Sales
select
    p.categoryname,
    sum(o.orderitemquantity * o.perunitprice) as total_sales
from product as p
join orderdetails as o
on p.productid = o.productid
group by p.categoryname
having total_sales > 50000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

categoryname	total_sales
CPU	5366747.48
Video Card	5746481.519999999
Storage	10712932.17
Mother Board	5555189.140000001
RAM	3138740.4

```sql
-- Q15 Display every warehouse along with the number of employees assigned to it, including warehouses that have no employees.
-- Display:
-- WarehouseID
-- WarehouseName
-- Employee Count
select
    w.warehouseid,
    w.warehousename,
    count(e.employeeid) as num_emp
from employee as e
right join warehouse as w
on e.warehouseid = w.warehouseid
group by w.warehouseid, w.warehousename;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

warehouseid	warehousename	num_emp
101	Southlake Texas	1
102	Southlake Texas	1
103	Southlake Texas	1
104	Southlake Texas	1
105	Southlake Texas	1
106	Southlake Texas	1

```sql
-- Q16 Find the total sales generated by each customer.
-- Display:
-- CustomerID
-- CustomerName
-- Total Sales
select
    c.customerid,
    c.customername,
    sum(o.orderitemquantity * o.perunitprice) as total_sales
from customer as c
left join orders on c.customerid = orders.customerid
left join orderdetails as o on orders.orderid = o.orderid
group by c.customerid, c.customername;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customerid	customername	total_sales
2301	Flor Stone	62038.68
2302	Lavera Emerson	64478.76
2303	Fern Head	73668.08
2304	Shyla Ortiz	108798.72
2305	Jeni Levy	11658.939999999999
2306	Matthias Hannah	81339.32
2307	Matthias Cruise	83818.73
2308	Meenakshi Mason	7588.62
2309	Christian Cage	93396.42

```sql
-- Q17 Find all products whose list price is greater than the average product list price.
-- Display:
-- ProductID
-- ProductName
-- CategoryName
-- ProductListPrice
select
    productid,
    productname,
    categoryname,
    productlistprice
from product
where productlistprice > (select  avg(productlistprice) from product);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

productid	productname	categoryname	productlistprice
P001	Intel Xeon E5-2699 V3 (OEM/Tray)	CPU	3410.46
P002	Intel Xeon E5-2697 V3	CPU	2774.98
P003	Intel Xeon E5-2698 V3 (OEM/Tray)	CPU	2660.72
P004	Intel Xeon E5-2697 V4	CPU	2554.99
P005	Intel Xeon E5-2685 V3 (OEM/Tray)	CPU	2501.69
P006	Intel Xeon E5-2695 V3 (OEM/Tray)	CPU	2431.95
P007	Intel Xeon E5-2697 V2	CPU	2377.09
P008	Intel Xeon E5-2695 V4	CPU	2269.99

```sql
-- Q18 Find all employees who work in warehouses located in the region with the highest number of warehouses.
-- Display:
-- EmployeeID
-- EmployeeName
-- WarehouseName
-- RegionID
select
    e.employeeid,
    e.employeename,
    w.warehousename,
    w.regionid
from employee as e
join warehouse as w
on e.warehouseid = w.warehouseid
where w.regionid = ( select regionid
                     from warehouse
                     group by regionid
                     order by count(regionid) desc limit 1);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employeeid	employeename	warehousename	regionid
6309	Summer Payne	Southlake Texas	1

```sql
-- Q19 Find the product categories whose total profit is greater than the average profit of all products.
-- Display:
-- CategoryName
-- Total Profit
select
    categoryname,
    sum(profit) as total_profit
from product
group by categoryname
having sum(profit) > ( select avg(profit) from product);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

categoryname	total_profit
CPU	24567.200000000004
Video Card	20036.15000000001
Storage	10888.060000000001
Mother Board	5735.070000000002
RAM	14671

```sql
-- Q20 Which five products generated the highest total sales revenue? Display the product ID, product name and total sales amount for each.
with highest_sales as (
select
    p.productid,
    p.productname,
    sum(o.orderitemquantity * o.perunitprice) as total_sales
from product as p
join orderdetails as o
on p.productid = o.productid
group by p.productid, p.productname
order by sum(o.orderitemquantity * o.perunitprice) desc
limit 5
)
select * from highest_sales;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

productid	productname	total_sales
P359	PNY SSD9SC240GMDA-RB	600155
P246	Samsung MZ-V6E500	550487
P304	Kingston Fury (HX316C10FB/8) 8 GB	389855.7
P273	Samsung MZ-75E120B/AM	344872.35
P267	Western Digital WD5000AACS	328038.42000000004

```sql
-- Q21 Find product categories whose total sales are higher than the average sales across all product categories.
-- Display:
-- CategoryName
-- Total Sales
-- Average Category Sales
with category_sales as (
    select
        p.categoryname,
        sum(od.orderitemquantity * od.perunitprice) as total_sales
    from product as p
    join orderdetails as od
    on p.productid = od.productid
    group by p.categoryname
)
select
    categoryname,total_sales,(select avg(total_sales) from category_sales) as Average_Category_Sales
from category_sales
where total_sales > (select avg(total_sales)from category_sales);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

categoryname	total_sales	Average_Category_Sales
Storage	10712932.17	6104018.142

```sql
-- Q22 Rank all products based on their profit margin, with the highest profit margin receiving rank 1.
-- Display:
-- ProductID
-- ProductName
-- CategoryName
-- Profit Margin
-- Profit Rank
select
    productid,
    productname,
    categoryname,
    profitmargin,
    dense_rank() over(order by profitmargin desc) as profit_rank
from (select
        productid,productname,categoryname,
        (profit / productlistprice) * 100 as profitmargin
    from product
) as p;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

productid	productname	categoryname	profitmargin	profit_rank
P291	Hynix (H15201504-8) Genuine DDR2 2 GB	RAM	40.08016032064128	1
P294	Simmtronics 2GB DDR2 667Mhz	RAM	33.47639484978541	2
P203	MSI X299 GAMING M7 ACK	Mother Board	29.870162548437417	3
P174	G.Skill Ripjaws V Series	Storage	29.86050836949783	4
P030	Intel Core i7-6950X	CPU	29.80018534692544	5

```sql
-- Q23 Rank products by their list price within each product category with the most expensive product receiving rank 1.
-- Display:
-- CategoryName
-- ProductID
-- ProductName
-- ProductListPrice
-- Category Rank
select
    categoryname,
    productid,
    productname,
    productlistprice,
    dense_rank() over(
        partition by categoryname
        order by productlistprice desc
    ) as category_rank
from product;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

categoryname	productid	productname	productlistprice	category_rank
CPU	P001	Intel Xeon E5-2699 V3 (OEM/Tray)	3410.46	1
CPU	P002	Intel Xeon E5-2697 V3	2774.98	2
CPU	P003	Intel Xeon E5-2698 V3 (OEM/Tray)	2660.72	3
CPU	P004	Intel Xeon E5-2697 V4	2554.99	4
CPU	P005	Intel Xeon E5-2685 V3 (OEM/Tray)	2501.69	5
CPU	P006	Intel Xeon E5-2695 V3 (OEM/Tray)	2431.95	6
CPU	P007	Intel Xeon E5-2697 V2	2377.09	7

```sql
-- Q24 Classify products based on their profit margin into three groups:
-- High Margin → 20% or above
-- Medium Margin → 10%–20%
-- Low Margin → below 10%
-- Display:
-- ProductID
-- ProductName
-- CategoryName
-- Profit Margin
-- Profit Category
select
    productid,
    productname,
    categoryname,
    (profit / productlistprice) * 100 as profitmargin,
    case
        when (profit / productlistprice) * 100 >= 20
            then 'High Margin'
        when (profit / productlistprice) * 100 >= 10
            then 'Medium Margin'
        else 'Low Margin'
    end as profit_category
from product;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

productid	productname	categoryname	profitmargin	profit_category
P001	Intel Xeon E5-2699 V3 (OEM/Tray)	CPU	15.920139805187572	Medium Margin
P002	Intel Xeon E5-2697 V3	CPU	16.1698462691623	Medium Margin
P003	Intel Xeon E5-2698 V3 (OEM/Tray)	CPU	23.510177696262666	High Margin
P004	Intel Xeon E5-2697 V4	CPU	16.070121605172623	Medium Margin
P005	Intel Xeon E5-2685 V3 (OEM/Tray)	CPU	19.56997069980693	Medium Margin
P006	Intel Xeon E5-2695 V3 (OEM/Tray)	CPU	20.840066613211622	High Margin
P007	Intel Xeon E5-2697 V2	CPU	11.589800975141875	Medium Margin
P008	Intel Xeon E5-2695 V4	CPU	21.57013907550254	High Margin

```
-- Q25 Create a product performance summary and use it to find the top 10 products based on total sales.
-- Display:
-- ProductID
-- ProductName
-- CategoryName
-- Total Quantity Sold
-- Total Sales
-- Total Profit
create view product_performance as
select
    p.productid,
    p.productname,
    p.categoryname,
    sum(od.orderitemquantity) as total_quantity_sold,
    sum(od.orderitemquantity * od.perunitprice) as total_sales,
    sum(od.orderitemquantity * p.profit) as total_profit
from product as p
join orderdetails od
on p.productid = od.productid
group by
    p.productid,
    p.productname,
    p.categoryname;

select
    productid,
    productname,
    categoryname,
    total_quantity_sold,
    total_sales,
    total_profit
from product_performance
order by total_sales desc
limit 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

productid	productname	categoryname	total_quantity_sold	total_sales	total_profit
P359	PNY SSD9SC240GMDA-RB	Storage	145	600155	0
P246	Samsung MZ-V6E500	Storage	133	550487	3242.54
P304	Kingston Fury (HX316C10FB/8) 8 GB	RAM	130	389855.7	26000
P273	Samsung MZ-75E120B/AM	Storage	115	344872.35	1676.7
P267	Western Digital WD5000AACS	Storage	138	328038.42000000004	0
P101	EVGA 11G-P4-6598-KR	Video Card	147	323400	21528.149999999998
P307	G.Skill RipjawsX (F3-1600C9Q-32GXM) DDR3 32...	RAM	149	307683.50999999995	412730
P156	G.Skill Trident Z	Storage	112	302398.88	13975.36
P213	Asus TUF X299 MARK 1	Mother Board	94	300139.18	9290.960000000001